

Sergio Junior Ambrosio Camayo

Civil Engineer — professional degree, Universidad Nacional de Ingeniería (UNI), Faculty of Civil Engineering

Structural Engineer | Steel, Concrete & Mining | Design Codes & Advanced Analysis | Python & Engineering Automation

Portfolio · LinkedIn · GitHub

Profile

Civil engineer specializing in steel and concrete structures for buildings, mining and electrical infrastructure. I combine design-code criteria, modelling and advanced analysis with structural design, verification and technical documentation. Python and engineering automation support results processing, parametric studies and consistency checks. I currently work on structural engineering for mining projects at SRK, with a focus on justified decisions and traceable deliverables.

Professional experience

SRK

Present

Structural engineering for mining projects

- Design of steel and concrete structures.
- Structural modelling, analysis and verification.
- Engineering work for mining infrastructure.

HMV Ingenieros

2026

Structural Engineer — Professional III

- Design of transmission towers, substation gantries and electrical equipment supports.
- Modelling and verification of steel structures and their foundations.
- Technical review of models, calculation reports, drawings and engineering deliverables.
- For CEPITOC, development of design criteria, assessment of design alternatives and technical coordination.

MRZ Ingenieros

November 2018 — January 2026

Structural Engineer

- Modelling and seismic analysis in ETABS of buildings with and without basements.
- Design and detailing of beams, columns, structural walls, slabs and foundations; foundation analysis in SAFE.
- Design of steel structures, industrial buildings and roof trusses.
- Assessment of jacketing and FRP strengthening solutions.
- Technical reviews, structural design optimization, calculation reports and drawings.
- Coordination of structural designs with architecture and MEP systems.

AMEC Foster Wheeler Perú

March — July 2018

Engineering Assistant

- Structural inspections.
- Topographic surveying.
- Preparation of inventories and quantity take-offs.

Grupo MIMCO S.A.C.

January 2017 — March 2018

Basic Engineering Assistant

- Contribution to the design of telecommunications towers and monopoles.
- Preparation of calculation reports.

Selected projects

SUNSETGOLF TORRE C · 2024

Structural modelling and design of a 21-storey residential tower with 3 basements, linking building analysis to member and foundation checks in coordination with architecture and MEP.

CEPITOCE / ELECTRICAL INFRASTRUCTURE · 2026

Design and verification of high-voltage towers, gantries and equipment supports at HMV Ingenieros, including steel design reviews, design criteria and technical documentation.

ALMACENES CIDELSA / YANACOCCHA · 2024

Design and verification of an approximately 20 × 30 m steel multipurpose tent structure, covering global analysis, members, connections and foundations, with Revit documentation and multidisciplinary coordination.

Research and publications

PERFORMANCE OF SEISMICALLY ISOLATED LIQUID STORAGE TANKS USING TRIPLE PENDULUM BEARINGS

S. J. Ambrosio; E. J. Mamani; V. I. Fernández-Dávila · 2021

17th World Conference on Earthquake Engineering · Sendai, Japan · 2021

PERFORMANCE OF SEISMICALLY ISOLATED WATER STORAGE TANKS

E. J. Mamani; S. Ambrosio; V. I. Fernández-Dávila · 2018

XXXVIII South American Conference on Structural Engineering · Lima · 2018

DESEMPEÑO SÍSMICO DE TANQUES PARA ALMACENAMIENTO DE AGUA CON BASE AISLADA

Sergio J. Ambrosio; Elvis J. Mamani; Víctor I. Fernández-Dávila · 2018

12th International Symposium on Disaster Risk Management · Lima · 2018

Expertise and tools

Reinforced concrete · Steel structures · Earthquake engineering · Nonlinear analysis · Finite element analysis · Computational engineering · Assessment and strengthening · Modelling and documentation

ETABS · SAFE · Mathcad · AutoCAD · SAP2000 · IDEA StatiCa · Revit · Tekla Structures · Python · OpenSeesPy · Abaqus · AutoLISP

Codes and verification: RNE E.020 / E.030 / E.050 / E.060; ACI 318, ASCE 7, AISC 360, ASCE 41, ACI 562. Scope and reference editions.

Python / AutoLISP: results processing, validation and documentation. New portfolio demonstrations (2026): CSV / SI · parametric cantilever.

Education

Civil Engineer — professional degree, Universidad Nacional de Ingeniería, Faculty of Civil Engineering.

Civil engineering studies: 2013–2017. Upper third of the class.